

ALGORITHMS FOR BLASTSMART APP

Step 1: Begin

Step 2: Input Blasting diameter (BD), Average In-Hole Density (p), Spacing to Burden ratio(a) and Bench Height (BH)

Step 3: Convert the inputs from String to Floating point numbers

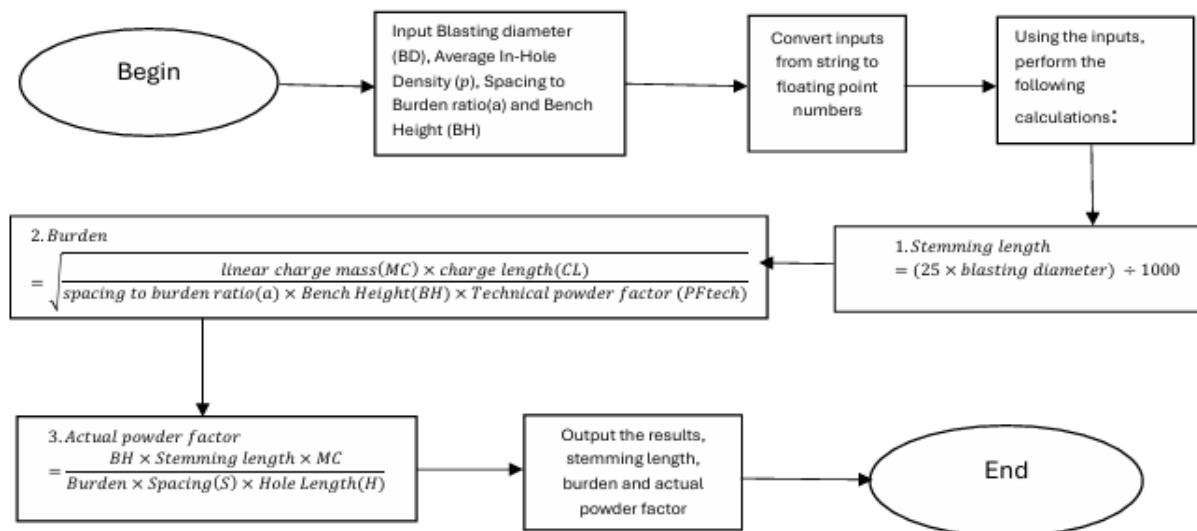
Step 4: Using the inputs, perform the following calculations:

1. *Stemming length* = $(25 \times \text{blasting diameter}) \div 1000$
2. *Burden* =
$$\sqrt{\frac{\text{linear charge mass}(MC) \times \text{charge length}(CL)}{\text{spacing to burden ratio}(a) \times \text{Bench Height}(BH) \times \text{Technical powder factor}(PFtech)}}$$
 - $MC = \frac{P \times BD^2}{1273}$
 - $CL = BH - \text{Stemming length}$ (assuming there is no sub – drill)
 - $PFtech = 0.3$ (standard $PFtech$ for limestone)
3. *Actual powder factor* =
$$\frac{BH \times \text{Stemming length} \times MC}{\text{Burden} \times \text{Spacing}(S) \times \text{Hole Length}(H)}$$
 - $\text{spacing} = a \times \text{Burden}$
 - $\text{Hole Length} = \text{Charge Length} + \text{Stemming Length}$

Step 5: Output the results, stemming length, burden and actual powder factor

Step 6: End

Flow Diagram



Algorithm 2: Fragmentation Calculator

Step1: Begin

Step2: Input Average in-hole density(p) , bench height(BH), Relative effective energy of explosives(E), Rock factor(A), Explosive type, and Stemming length(SL)

Step3: Convert all the inputs from Strings to Floating point numbers

Step4: Check if Explosive type is ANFO, then assign E to 100

 :Check if Explosive type is TNT, then assign E to 115

 : Otherwise assign E to 150

Step5: Substitute the inputs to the equation below to calculate for the mean fragment size

1. $Q = BH - SL$
2. Mean fragment size = $A * (K^{**(-0.8)}) * (Q^{**(1/6)}) * ((115/E)^{** (19/20)})$

Step6: Output Q

Step7: Output Mean fragmentation size

Step8: End

